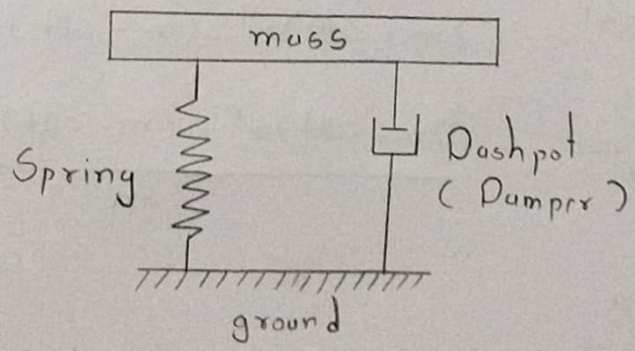
# Numerical Simulation of Undrody Vibrations in a Mass-Spring-Dumper Sytem using FDM

* Objective
  - To analyze the dynamic response of a vehicle suspension system.
  - To solve the governing equation using Finite Difference Method
  - To study underdamped, critically dumped, and overdamped cases.

* Vehicle Suspension system : A suspension system consists of
  - mass ( 1/4 of vehicle )
  - Spring (suspension )
  - Dumper ( shock absorber )

  → It is used to Absorb road shocks, Provide comfort, Maintain stability.




* Governing Equation

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + Kx = 0$$

$m$ = mass
$c$ = damping coefficient
$K$ = stiffness
$x(t)$ = displacement

* Finite Difference Method (FDM)

- Second derivative

$$\frac{d^2x}{dt^2} = \frac{x^{n+1} - 2x^n + x^{n-1}}{\Delta t^2}$$

- First derivative

$$\frac{dx}{dt} = \frac{x^n - x^{n-1}}{\Delta t}$$

$$\rightarrow \quad m\left(\frac{x^{n+1} - 2x^n + x^{n-1}}{\Delta t^2}\right) + c\left(\frac{x^n - x^{n-1}}{\Delta t}\right) + Kx^n = 0$$

$$m(x^{n+1} - 2x^n + x^{n-1}) + c\Delta t(x^n - x^{n-1}) + K\Delta t^2 x^n = 0$$

$$mx^{n+1} - 2mx^n + mx^{n-1} + c\Delta t x^n - c\Delta t x^{n-1} + K\Delta t^2 x^n = 0$$

$$mx^{n+1} = (2m - c\Delta t)x^n - (m - c\Delta t)x^{n-1} - K\Delta t^2 x^n$$

$$x^{n+1} = \frac{(2m - c\Delta t)x^n - (m - c\Delta t)x^{n-1} - K\Delta t^2 x^n}{m}$$

→ Given Data

$m = 250$ Kg
$K = 20000$ N/m
$\Delta t = 0.01$ s
$x_0 = 0.05$m = Initial displacement
$V_0 = 0$ = Initial velocity

→ Initial Conditions

$x^1 = x^0 + v^0 \Delta t$

→ Conditions

$C_c = 2\sqrt{mK}$ (Critical Damping)
$c = 0.5\, C_c$ (Underdamped)
$c = 2C_c$ (Overdamped)